

MLPC – complementary slides

Team Quantized Encoders

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Metadata

Diverse set of metadata labels, including:

- filename
- target_classes
- recording_device
- device_placement
- recording_environment
- scene_description
- license
- collector_id
- non_target_classes

Main Focus on environments & device placement

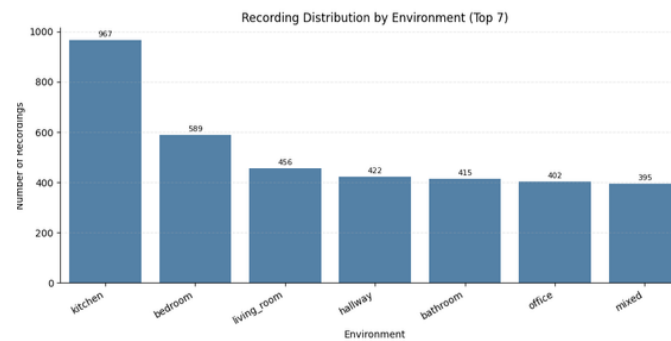


Figure 5: Recording Distribution by Environments, Top 7

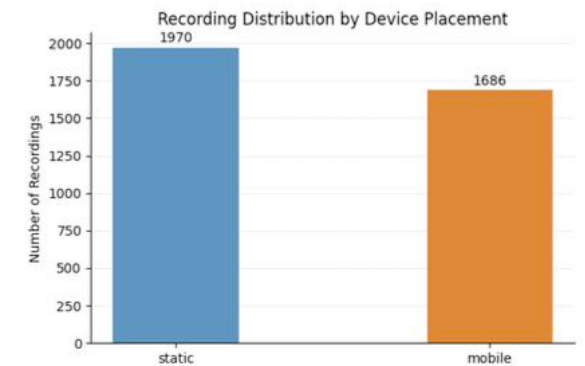


Figure 6: Recording Distribution by Device Placement

Feature Statistics & Correlation

- Strong scale differences across features
- High variance in energy-related features
- Higher stability in spectral features
- Low variance features

→ Complementary properties,
but demands preprocessing

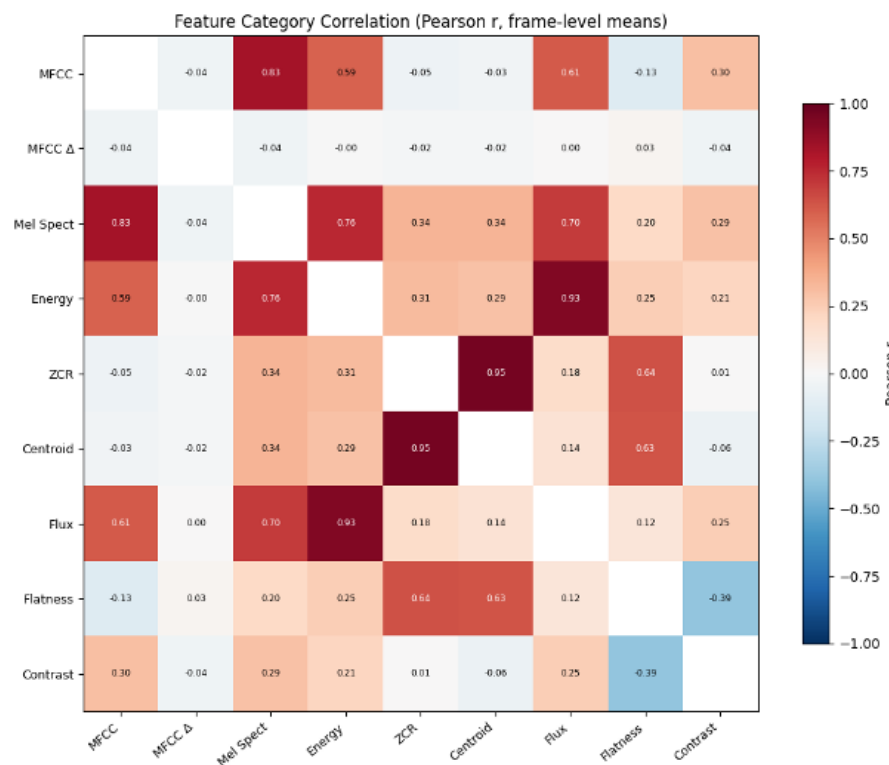


Table 1: Feature Statistics

Feature	Mean	Std	Range
ZCR	0.192	0.093	0.943
Energy	31.178	32.744	1978.073
Power	62.717	121.911	67940.875
Spectral Flux	1.929	1.703	166.599
Flatness	0.088	0.062	1.000
Centroid	2208.360	632.617	7615.978
Bandwidth	2785.426	377.676	4951.661
Rolloff Low	300.506	210.128	7437.500
Rolloff High	5074.285	960.228	7937.500
MFCC	-2.051	0.391	169.512
MFCC delta	0.000	0.000	32.648
MFCC delta2	-0.000	0.000	18.633
Mel Spectrogram	4.494	1.169	13.180
Contrast	15.737	0.928	80.107

Feature Space Visualizations

- Dense, highly overlapping distribution
 - Weak tendency for partial clusters (microwave, bell, keychain,...)
 - Frame-level features indicate lack of temporal context
- Without thorough preprocessing only limited separability in feature space given

